

THEME 2: LIMITS, CONTINUITY AND DERIVATIVES

Unit 2.1 The tangent and velocity problems

Source: Stewart, Section 2.1 pp 78 - 82.

Learning objectives

On completion of this unit you should

1. be able to estimate the slope of a tangent line to a curve by calculation of the limit of the slopes of secant lines.
2. be able to explain the concepts "average velocity over an interval" and "instantaneous velocity at time t ".
3. be able to estimate the instantaneous velocity at time t by finding the average velocity over intervals if a distance function is given.

Remark

At a later stage we will develop easier ways of finding instantaneous velocity and the slopes of tangent lines.

Assigned Problems

Stewart, Exercises 2.1 p 82: Numbers 1(a), 1(b), 8.

Unit 2.2 The limit of a function

Source: Stewart, Section 2.2 pp 83 - 94.

Learning objectives

On completion of this unit you should

1. be able to find the limit of a function at a point if a graph is given.
2. be able to explain what is meant by one-sided limits and be able to interpret these ideas graphically.
3. be able to find $\lim_{x \rightarrow a} f(x)$ by using the one-sided limits $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ (when necessary).
4. be able to explain the informal definition of an infinite limit ($\lim_{x \rightarrow a} f(x) = \infty$ or $\lim_{x \rightarrow a} f(x) = -\infty$) and be able to interpret this idea graphically.
5. be able to find vertical asymptotes of a function.

Remarks

1. It is important to understand the informal idea of a limit as given on p 83. The formal precise definition is given on p 106. You do not have to know this definition.

- Examples 3 and 5 are important, they show why we need better methods to find limits.
- In order to find the vertical asymptotes of a function: Find the domain of the function. If $x = c$ is in the domain of the function then $x = c$ is NOT a vertical asymptote. If the domain is (a, ∞) , calculate $\lim_{x \rightarrow a^+} f(x)$. If the domain is $(-\infty, a)$, calculate $\lim_{x \rightarrow a^-} f(x)$. If the domain is (a, b) , calculate $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$.

Assigned Problems

Stewart, Exercises 2.2 p 92: Numbers 1, 2, 5, 7, 9, 11, 15, 17, 29, 32, 33, 35, 36.

Unit 2.3 Limit laws

Source: Stewart, Section 2.3 pp 94 - 104.

Learning objectives

On completion of this unit you should

- be able to use the appropriate limit law to find
 - the limit of the sum of two functions;
 - the limit of the difference of two functions;
 - the limit of a constant multiple of a function;
 - the limit of the product of two functions;
 - the limit of the quotient of two functions;
- be able to use the direct substitution property to find the limit of polynomials and rational functions;
- be able to determine whether a limit exists by looking at the one-sided limits at the point;
- be able to write down and use the squeeze theorem;
- be able to use the limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, to determine similar limits, e.g., $\lim_{x \rightarrow 0} \frac{x}{\sin x}$, $\lim_{x \rightarrow 0} \frac{\tan x}{x}$ and $\lim_{x \rightarrow 0} \frac{x}{\tan x}$.

Remarks

- These laws can be proved by using the formal definition of a limit.
- The method used in Example 6 is important.
- Omit Example 10 on p 101.

4. The “proof” that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ appears on p 195 but you do not have to know it.

Assigned Problems

1. Stewart, Exercises 2.3 p 102: Numbers 2, 13, 15, 21, 23, 27, 29, 41, 46, 47, 52, 61, 64.
2. Stewart, Exercises 3.3 p 197: Numbers 45, 46, 53, 54, 55.

Unit 2.4 Continuity

Source: Stewart, Section 2.5 pp 115 - 126.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of continuity of a function at a point;
2. be able to explain the definition of continuity at a point graphically;
3. be able to determine if a given function is continuous at a point;
4. be able to distinguish between removable, jump and infinite discontinuities;
5. be able to write down and use the definitions of “continuous from the left” and “continuous from the right”;
6. be able to test whether a function is continuous over an interval. You have to distinguish between the intervals $[a, b]$, (a, b) , $(a, b]$, $[a, b)$, etc.;
7. be able to use the theorems on the continuity of combinations of functions (sum, difference, constant multiple, products, quotient);
8. know that the following functions are continuous in every point of their domains
 - polynomials · rational functions · root functions
 - exponential functions · logarithmic functions · trigonometric functions
 - inverse trigonometric functions
9. know that the composite of continuous functions is continuous and be able to use this fact to calculate limits.

Remarks

1. Omit the proofs of the theorems and the Intermediate Value Theorem.
2. Omit Examples 2(d), 3 and 10.

Assigned Problems

Stewart, Exercises 2.5 p 124: Numbers 3, 5, 7, 9, 13, 20, 23, 41, 43, 48, 49.

Unit 2.5 Limits at infinity and horizontal asymptotes

Source: Stewart, Section 2.6 pp 127 - 140.

Learning objectives

On completion of this unit you should

1. be able to explain what is meant by the following limits:

$$\lim_{x \rightarrow \infty} f(x) = L, \quad \lim_{x \rightarrow -\infty} f(x) = L, \quad \lim_{x \rightarrow \infty} f(x) = \infty, \quad \lim_{x \rightarrow -\infty} f(x) = \infty,$$

$$\lim_{x \rightarrow \infty} f(x) = -\infty, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

and to interpret the limits graphically.

2. be able to find horizontal asymptotes of a function.
3. be able to find $\lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)}$, with $f(x)$ and $g(x)$ polynomials by using the fact that $\lim_{x \rightarrow \pm\infty} \frac{1}{x^r} = 0$, for r a positive rational number.

Remarks

1. You have to memorize the two limits given in [4] on p 132.
2. Example 4 shows that there may be more than one horizontal asymptote.
3. Omit Example 12. We will sketch functions later.
4. Omit the precise definitions and examples on pp 134 - 137.

Assigned Problems

Stewart, Exercises 2.6 p 137: Numbers 3, 5, 8, 15, 17, 19, 23, 27, 33, 37, 41, 47, 49, 51, 52, 56.

Unit 2.6 Derivatives and rates of change

Source: Stewart, Section 2.7 pp 140 - 152.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the tangent line to a curve.
2. be able to find the equation of the tangent line to a graph.
3. be able to interpret average velocity and instantaneous velocity as slopes of graphs.

4. be able to find the (instantaneous) velocity of an object if the distance is given as a function of the time.
5. be able to write down and use the definition of the derivative of a function at a point.
6. know that the derivative is the slope of the tangent line to the curve.
7. be able to find derivatives of functions by using the definition.
8. be able to estimate derivatives from tabled data.
9. be able to distinguish between average rate of change over an interval and (instantaneous) rate of change at a point.

Remarks

1. There are two formulas to calculate the derivative at a point, [4] and [5] on p 144. You have to know both.
2. Omit Examples 7 and 8.

Assigned Problems

Exercises 2.7 p 149: Numbers 3, 7, 12, 16, 17, 18, 19, 28, 34, 39, 43, 45, 57.

Unit 2.7 The derivative as a function

Source: Stewart, Section 2.8 pp 153 - 166.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the derivative function;
2. be able to find the formula of the derivative function by using the definition;
3. be able to sketch the graph of the derivative function if the graph of the function is given;
4. know the other notations for the derivative;
5. be able to write down and use the definition of a function that is differentiable at a point;
6. be able to write down and use the definition of a function that is differentiable on an interval;
7. know and be able to prove that a differentiable function is continuous in a point and that the converse of the theorem is not true;

8. know the three ways for a function not to be differentiable at a point;
9. be able to illustrate graphically when a function is differentiable in a point;
10. be able to illustrate graphically when a function is not differentiable in a point;
11. know the notation for higher derivatives.

Remarks

1. You have to be able to prove Theorem 4 on p 157.
2. Omit Examples 6 and 7.

Assigned Problems

Stewart, Exercises 2.8 p 161: Numbers 3, 4, 9, 23, 27, 41, 42, 47, 49, 51, 63.